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Confined enclosure insect capture device with adhesive gap

Abstract

An insect capture device is used to monitor, capture, and/or kill insects. The device has a raised portion to create a gap between a surface and the device to entice insects to pass under. At the raised portion, a glue board includes an adhesive facing the surface. As the insect passes under the raised portion, the insect can unknowingly become attached to the adhesive portion and stuck thereto. To aid in installing the glue board, one or more sections of the glue board can be free of adhesive to connect or fit to a part of the device to hold the glue board in place, while also allowing for easy removal to replace the glue board with a new one, free of captured insects.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority under 35 U.S.C. § 119 to U.S. Provisional Application Ser. No. 62/706,671, filed Sep. 2, 2020. The provisional patent application is herein incorporated by reference in its entirety, including without limitation, the specification, claims, and abstract, as well as any figures, tables, appendices, or drawings thereof.

FIELD OF THE INVENTION
(1) The invention relates generally to an insect pest monitor and/or product transfer station. More particularly, but not exclusively, the invention relates to a device including a glue board wherein the glue board includes various sections of glue and non-glue to allow for easy and accurate installation of the glue board with the insect capture device.
BACKGROUND OF THE INVENTION
(2) The background description provided herein gives context for the present disclosure. Work of the presently named inventors, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art.
(3) Crawling pests, such as the German cockroach (*Blattella germanica*) are well known to carry disease and are widely considered to be undesirable insects. The German cockroach is a smaller member of the cockroach family and is frequently a pest in food processing and preparation areas

including hotels, nursing homes, and other institutions. They are widespread pests capable of surviving in many different parts of the world. They are a positively thigmotactic insect, meaning that they generally react to a physical stimulus, and therefore prefer tight spaces. Such insects frequently hide out of sight in cracks and crevices that are easy for humans to overlook.
(4) Additional insects, including flying insects, such as Indian meal moths and other types of moths, may also be enticed to tight spaces.
(5) Numerous designs of insect pest stations are commercially available, some use large containment areas, while others use wide open, flat surfaces with various forms of attractants and capture mechanisms, such as glue boards. Problems exist with such devices. For example, cockroaches have been observed to contact the edges of glue boards and escape. It is known that when such glue boards are rolled into cylinders, cockroaches would fill the underside of the glue roll. Similarly, a flat glue board can be placed with its glue side or sticky side down and another flat glue board was placed with its sticky side up. It has been found that the entire surface of the glue side down board was filled while only the edges of the sticky side up board were filled. Thus, it is desirable to provide an insect monitor that is directed to insets who desire small, narrow, or covered spaces.
(6) Insect devices exist wherein an adhesive is placed relative to the device so an insect moves generally underneath the adhesive and gets “caught” by the overhead adhesive. For example, U.S. Pat. Nos. 10,123,522 and 10,412,952, which are co-owned and incorporated by reference in their entirety, disclose a station that has a cutout wherein an adhesive is positioned through the cutout and faced generally downward to capture insects.
(7) However, in order to improve upon that which exists, there is a need in the art for a combination of a capture device and adhesive in which the adhesive is easily and accurately positioned relative to the capture device to ensure the best possible capturing of insects.
SUMMARY OF THE INVENTION
(8) The following objects, features, advantages, aspects, and/or embodiments, are not exhaustive and do not limit the overall disclosure. No single embodiment need provide each and every object, feature, or advantage. Any of the objects, features, advantages, aspects, and/or embodiments disclosed herein can be integrated with one another, either in full or in part, even if not explicitly disclosed.
(9) It is a primary object, feature, and/or advantage of the invention to improve on or overcome the deficiencies in the art.
(10) It is a further object, feature, and/or advantage to provide a combination of a device and glue board to capture insects passing thereby.
(11) It is still yet a further object, feature, and/or advantage to provide a glue board that can be easily and accurately installed in an insect capture device.
(12) It is another object, feature, and/or advantage to have a combination glue board and capture device wherein the glue board is positioned overhead of an area in which an insect will pass.
(13) It is still another object, feature, and/or advantage to provide a method of installing a glue board into a capture device that allows for quick, easy, and full exposure of the adhesive portion of the glue board.
(14) It is yet another object, feature, and/or advantage to provide a glue board having an adhesive area, one or more non-adhesive areas, and a cover paper to allow for installation of the glue board.
(15) Aspects of the disclosure, including the glue board disclosed herein can be used in a wide variety of applications. For example, it is to be appreciated that it could be used as a standalone device. In addition, it could be used with a number of housings or devices for mounting and capturing insects, including those not explicitly disclosed herein.
(16) It is preferred the apparatus be safe, cost effective, and durable.
(17) At least one embodiment disclosed herein comprises a distinct aesthetic appearance. Ornamental aspects included in such an embodiment can help capture a consumer's attention and/or

identify a source of origin of a product being sold. Said ornamental aspects will not impede functionality of the invention.
(18) Methods can be practiced which facilitate use, manufacture, assembly, maintenance, and repair of an insect device which accomplish some or all of the previously stated objectives.
(19) The apparatus, combination, or portions thereof can be incorporated into systems or kits which accomplish some or all of the previously stated objectives.
(20) According to some aspects of the present disclosure, [first independent claim **1**].
(21) According to some additional aspects of the present disclosure, [dependent claims].
(22) According to some other aspects of the present disclosure, [subsequent independent claim].
(23) These and/or other objects, features, advantages, aspects, and/or embodiments will become apparent to those skilled in the art after reviewing the following brief and detailed descriptions of the drawings. Furthermore, the present disclosure encompasses aspects and/or embodiments not expressly disclosed but which can be understood from a reading of the present disclosure, including at least: (a) combinations of disclosed aspects and/or embodiments and/or (b) reasonable modifications not shown or described.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Several embodiments in which the invention can be practiced are illustrated and described in detail, wherein like reference characters represent like components throughout the several views. The drawings are presented for exemplary purposes and may not be to scale unless otherwise indicated.
(2) FIG. **1** is a perspective view of an insect capture device or station according to aspects and embodiments disclosed herein.
(3) FIG. **2** is a side elevation view of the insect capture device.
(4) FIG. **3** is a top plan view of the insect capture device.
(5) FIG. **4** is a bottom plan view of the insect capture device.
(6) FIG. **5** is an exploded view of the insect capture device.
(7) FIG. **6** is an exploded, bottom view of the insect capture device.
(8) FIG. **7** is an exploded view showing additional aspects of a housing and removal key in exploded form.
(9) FIG. **8** is a bottom, plan view of the housing and removal key.
(10) FIG. **9** is a perspective view of the housing of an insect capture device according to embodiments and/or aspects disclosed herein.
(11) FIG. **10** is a side elevation view of the housing.
(12) FIG. **11** is a bottom, plan view of the housing.
(13) FIG. **12** is a perspective view of a base member of an insect capture device according to embodiments and/or aspects disclosed herein.
(14) FIG. **13** is a perspective view of a glue board for use with an insect capture device according to embodiments and/or aspects disclosed herein.
(15) FIG. **14** is a side elevation view of the glue board.
(16) FIG. **15** is an enlarged, side elevation view of the housing and glue board.
(17) FIGS. **16**A-**16**D show depictions of the install of an exemplary insect capture device and removal of the protective cover from a glue board.
(18) FIG. **17** is a schematic of a production glue board for use with the embodiments and/or aspects disclosed.
(19) An artisan of ordinary skill need not view, within isolated figure(s), the near infinite number of distinct permutations of features described in the following detailed description to facilitate an

understanding of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(20) The present disclosure is not to be limited to that described herein. Mechanical, electrical, chemical, procedural, and/or other changes can be made without departing from the spirit and scope of the invention. No features shown or described are essential to permit basic operation of the invention unless otherwise indicated.

(21) Unless defined otherwise, all technical and scientific terms used above have the same meaning as commonly understood by one of ordinary skill in the art to which embodiments of the invention pertain.

(22) The terms “a,” “an,” and “the” include both singular and plural referents.

(23) The term “or” is synonymous with “and/or” and means any one member or combination of members of a particular list.

(24) The terms “invention” or “present invention” are not intended to refer to any single embodiment of the particular invention but encompass all possible embodiments as described in the specification and the claims.

(25) The term “about” as used herein refer to slight variations in numerical quantities with respect to any quantifiable variable. Inadvertent error can occur, for example, through use of typical measuring techniques or equipment or from differences in the manufacture, source, or purity of components.

(26) The term “substantially” refers to a great or significant extent. “Substantially” can thus refer to a plurality, majority, and/or a supermajority of said quantifiable variable, given proper context.

(27) The term “generally” encompasses both “about” and “substantially.”

(28) The term “configured” describes structure capable of performing a task or adopting a particular configuration. The term “configured” can be used interchangeably with other similar phrases, such as constructed, arranged, adapted, manufactured, and the like.

(29) Terms characterizing sequential order, a position, and/or an orientation are not limiting and are only referenced according to the views presented.

(30) The “scope” of the invention is defined by the appended claims, along with the full scope of equivalents to which such claims are entitled. The scope of the invention is further qualified as including any possible modification to any of the aspects and/or embodiments disclosed herein which would result in other embodiments, combinations, subcombinations, or the like that would be obvious to those skilled in the art.

(31) Exemplary features and aspects of the invention for monitoring, attracting, and killing insects, such as cockroaches, beetles, Indian meal moths, etc., within a short window of time are illustrated in FIGS. **1-6**. As shown in the exemplary embodiments of FIGS. **1-6**, the insect capture device **10**, which also may be referred to as an insect station, system, capture station, monitoring station, trap or the like, includes generally a housing **12**, a base **40**, and an adhesive member **50**. It should initially be noted that not all of the elements be required in all embodiments. For example, as will be understood, while the base **40** can be utilized to aid in positioning and/or temporarily attaching the device **10** to a surface, it need not be required. Instead, the housing **12** as is shown in the figures, or in a slightly modified manner, can be positioned, affixed, or otherwise used with a surface without the use of a base **40**, such as with an adhesive, mechanical fastener, or other means. Therefore, the invention should not be limited to the explicit embodiments as shown.

(32) Referring back to the figures, the insect capturing device **10** has a housing **12**, which includes a raised portion **22** to create a gap **24** between the raised portion **22** and a surface for use of the device **10**. It is envisioned that the device **10** could be used with any number of surfaces, including floors, walls, crawl spaces, interior, exterior, and the like. As will be appreciated, the raised portion **22** of the housing **12** creating said gap **24** will provide an enticing area for insects to enter. The combination of the raised portion **22** and an adhesive member **50**, such as a glue board, as will be understood, will aid in trapping said insect with the device **10**. Still further, a pesticide or other

killing agent could then be used to kill the captured insects. According to some embodiments, the pesticide could be used to kill any insect that is not caught or captured by the adhesive, and the captured insects could be killed by the pesticide or otherwise. In addition, it is contemplated that a pesticide and/or bait could be used without the adhesive glue board. The device **10** provides a harboring area for the insects, which are then exposed to the pesticide/bait, who could then return to their colony to expose more insects.

(33) As disclosed herein and also in co-owned U.S. Pat. Nos. 10,123,522, and 10,412,952, which are incorporated by reference in their entirety, having an insect device with a raised portion to create a gap is enticing to insects. Furthermore, the use of an adhesive member in an upper portion of the gap in order to trap insects, with or without wings, has been shown to be more effective and efficient in the monitoring and capturing of insects. The insects are able to pass into the gap without realizing the location of the adhesive member and become affixed thereto, upon which time it is too late for them to continue on. Still further, it should be appreciated that pesticides could also be used in conjunction with the adhesive member such as on the adhesive member or adjacent the adhesive member such as on the surface or surrounding areas to further aid in capturing and killing insects with the insect device **10**.

(34) Referring now to FIGS. **9-11**, exemplary housing **12** for use with the insect capture device **10** is shown. The housing **12** comprises a member having an upper or outer side **14** and an opposite or inner bottom side **16**. The terms outer and inner refer generally to the use of the device and in conjunction with the insects. For example, the inner side **16** of the housing **12** will generally be facing the surface on which the device **10** is placed, while the outer or upper side **14** is positioned opposite and away from the surface for use. At or in conjunction with the inner side **16** are a first and second foot or support members/surfaces **18**, **20**. These components of the housing **12** are used as a portion of the housing **12** to be positioned on the support surface. For example, when the surface is a floor, the housing **12** will be positioned and supported via the feet **18**, **20** on the surface. In addition, as noted, the housing **12** includes a raised portion **22** at the inner side **16** of the housing **12**. The term raised refers to the fact that there is a gap **24** positioned between generally at least one of the feet **18**, **20** or the surface **62** and the inner wall **16** of the housing. This is shown best in FIG. **10**, where the gap **24** is shown via the width between the surface **62** and the raised portion **22**.

(35) As shown in FIG. **10**, the raised portion and generally the inner section **16** is an arched or curved shape spanning generally from a first support foot **18** to the second support foot **20**. The arched, curved, or dome shape is enticing to insects and will aid in attracting the insects to pass thereunder. However, it should be appreciated that the dome, curve, or arched shape is not needed in all embodiments. For example, the housing **14** could include generally right angles that have a raised section **22** with a gap that has a rectangular, square, or other geometrical shape when looking from the side, such as in FIG. **10**. Furthermore, the height of the gap **24** from generally the surface **62** to the raised portion **22** can vary depending on the type of insect to be attracted and captured by the device **10**. Therefore, while a curved and dome shape is shown, it should be appreciated that the invention is not limited to such a shape, and that generally any housing having a raised portion in which there is a gap **24** between a lower portion of the housing, such as a support or foot and the raised portion **22** to entice and having a mounting surface for an adhesive member should be considered to be included as part of the present disclosure.

(36) Also shown in the figures for the housing **12** are support ribs **26**. The support ribs **26** are provided as both support for an adhesive member, also to provide support to the housing such that it is able to support itself and maintain structural integrity to the use and lifespan of the device. The exact number and type of support ribs **26** are not to be limiting on the invention.

(37) Shown best in FIG. **10** are a first notch or ledge **28** and a second notch or ledge **30**. The first notch **28** is positioned generally at or near the first support member **18** and positioned on the inner section **16** of the housing **12**. It is also noted that the first notch **28** is positioned generally at a beginning of the raised portion **22**, such as at the rear of the housing **12**. The second notch or ledge

30 is positioned opposite the first notch or ledge **28**, and is generally positioned at or near the second support member **20** of the housing **12**. It should be noted that the notches **28**, **30** are positioned on generally opposite ends of the raised inner section and spanned substantially the length of the raised inner section **22**. As will be understood, the notches or ledges **28**, **30** are used to install and support an adhesive member, such as a glue board **50**, as will be explained herein. While the components are referred to as a notch or ledge, it should be appreciated that these are not to be limiting. For example, the notches or ledges **28** are used to hold an adhesive member, such as a glue board, as has been disclosed. Therefore, the width of the notches, as well as the exact configuration, shape, and/or type of ledge can be varied, such as according to the type of adhesive or glue board, including the shape, thickness, and/or type of glue or adhesive to be used with the housing **12** of the device **10**. Furthermore, while both a first and second notch or ledge **28**, **30** are shown in the figures, it should be appreciated that they may not need both to be required in both instances. For example, it is envisioned that only one notch or ledge be used to support the glue board or adhesive with the device **10**. Still further, as best shown in FIG. **11**, the notches or ledges do not span the entire width of the housing **12**, but could be varied to span said full length or generally at any portion of the width of the housing.

(38) However, having both the first and second ledges **28**, **30** will aid in conforming an adhesive, such as a glue board **50**, generally to the shape of the raised portion **22** of the housing **12**. For example, when a curved shape for the raised portion is used, such as is shown in the figures, the first and second ledges **28**, **30** will aid in conforming the glue board to said curved shape, which will enhance the usage by making sure that the glue board is substantially close to the inner surface **16** along the raised portion, which has been designed to entice and capture insects of varying sizes. Still further, it should be appreciated that the ledges or notches **28** are shown to be generally flange like members protruding out from the inner portion **16** or the housing **12**. However, it should also be appreciated that the notches **28**, **30** could be formed such as pockets or other apertures extending into a portion an inner wall **16** and not need to be protruding outward therefrom. In any aspect, the notches or ledges **28**, **30** will be used to aid and holding an adhesive, pesticide, or other board like member for use with the housing **12**.

(39) Additional components of the housing **12** include protrusions **32** in the form of snap like members. The protrusions **32** can be used when the housing **12** will be used with a base member **40**. The snap like protrusion **32** can extend into a portion of the base to temporarily hold the housing **12** in connection with the base member **40**. This will allow for placement of the housing **12** relative to the base, but will also provide for an easy to remove feature and removal of the housing relative to the base **40**, such as for a replacement of components, or even monitoring of the station **10**. The housing **12** includes a face portion **34**, which will generally mate with the base potion **40**, such as at the base **42** thereof, in order to connect the pieces together.

(40) Still further, as best shown in FIG. **12**, the base **40** has apertures **44** that will allow for the insertion of the snap members **32** of the housing **12** to be inserted therein to connect, at least temporarily, the housing **12** to the base portion **40**. The base **40**, as shown in FIG. **12**, includes a mounting surface **46**. The mounting surface is an area which can be mounted to a surface, such as surface **62**. The base **40** can be connected or positioned on the surface **62** in a number of ways, including mechanical connecting means, adhesives, resting or otherwise simply placing, or generally any number of ways to affix a permanent or non-permanent manner.

(41) Referring back to FIG. **9**, additional components of the housing **12** are shown with respect to the outer or upper side **14**. For example, the upper or outer side may include a protrusion **38** in the form of a lifting fin or other fin like member. The protrusion **38** can be used to aid in lifting the housing at a location. Still further, the outer portion can include a section for a label **39**. The label can be in the form of a sticker, paint, mold, or the like. For example, the label section **39** can be used to provide some sort of information for use of the device **10**. As shown in FIG. **9**, the label **39** includes the words “poison” and “do not touch”, in both English and a foreign language. It is

envisioned that the label could include the exact type of poison or other chemicals that are used with the device in order to provide information to a user of the device. In addition, it should be appreciated that unit will need not be required in all embodiments.

(42) As disclosed, the device **10** using the housing **12** can be used with an adhesive device to capture insects, such as a glue board **50**. In addition to that which has been disclosed, the glue board can be positioned relative to portions of the inner side **16** of the housing **12**, such as the notches **28**, **30**, in order to conform the shape of the glue board **50** to match or substantially match the of the raised portion of the housing **12**. This is shown in FIGS. **1** and **2**. As shown in FIGS. **1** and **2**, the glue board **50** is positioned at the underside or inner portion **16** of the housing **12**. Furthermore, the glue board has a base portion **52** which can be positioned generally within the notches or ledges **28**, **30**, which aid in holding the glue board in place relative to the housing **12**. Still further, as the notches are on the opposite sides of the raised portion **22** of the housing **12**, the notches **10** urge the glue board **50** to take generally the shape of the raised portion **22**. For example, as shown in FIG. **2**, the raised portion **22** is in a curved or arched shape. Therefore, the glue board has been urged to be in the same or similar arched or curved shape of the raised portion **22** as well. The notches and/or ledges **28**, **30** aid in holding the glue board in place at the inner portion **16** of the housing **12**.

(43) However, when the raised portion takes additional or alternative shapes, including generally planar parallel shapes to that of the ground, or having more squared off sections, the notches or ledges can still hold the glue board **50** in place relative to the housing to aid in capturing insects. For example, it is envisioned that the raised portion **22** be generally parallel to the surface of the location of use, such that the raised portion **22** will be substantially parallel and possibly planar as well. It could have one singular height spanning from one ledge **28** to the other 30. In such situation, the glue board may conform to the general planar shape and be generally parallel to the surface of use, such that the height of the glue board be substantially the same from one end to the other. Still further, and as will be understood, the glue board includes an adhesive portion. The adhesive portion is positioned on the base **52** and is positioned to be exposed away from the housing **12** and towards the surface **62**. Therefore, when the glue board is positioned similar to that shown in FIG. **2**, the portion of the glue board **50** adjacent the inner wall **16** of the housing **12** may or may not have an adhesive, while the opposite side of the adhesive facing the surface **62** and positioned generally away from the housing **12** can have an adhesive side. Such an adhesive can be used to catch or trap the insects passing thereby, such as by the wings or upper portion thereof or even feet thereof. However, it should also be appreciated that the glue board **50** include adhesive on both sides thereof. This would allow the glue board to be affixed, at least temporarily to the inner wall **16** of the housing **12**, having both sides have an adhesive can aid in the conforming of the shape of the glue board to that of the raised portion **22** of the housing **12**, while still having an adhesive exposed to the gap below the inner wall **16** of the housing **12** to be able to catch and capture insects passing therethrough. In such a situation, it is to be appreciated that the adhesives need not be the same. For example, the adhesive on the base portion attached to the housing wall **16** could have less adhesive qualities than that of the insect side. This would allow for the easy installation and removal of the glue board **50** relative to the housing **12**, while maintaining a strong adhesive for capturing and catching of any insect passing therethrough.

(44) FIGS. **5** and **6** are exploded views of the housing **12**, base **40**, and glue board **50** of the insect capture device **10**. The exploded views can aid showing how the components can be connected to one another. For example, as shown in FIG. **5**, the housing and base can be moved along the arrows **78** in the direction of the arrows **78** to insert the snapped **32** to the aperture **44** of the base in order to affix, at least temporarily, the housing **12** to the base **40**. In addition, while not installed in such a manner, for exemplary purposes, the glue board **50** can be moved via the arrows **80** in FIGS. **5** and **6** which would show the placements of the glue board **50** at the inner walls **16** of the housing **12**, and nestled generally in the notches or ledges **30** thereof. Therefore, FIGS. **5** and **6** should not be

limiting but should be used to show how the components can be positioned relative to one another to be used as an insect capture device **10**.

(45) An additional feature of the insect device **10** is shown in FIGS. **7** and **8**. As shown in FIGS. **7** and **8**, the housing **12** is shown. On the connecting face **34** of the housing **12**, there are shown to be apertures **36** and a dividing wall **37**. It should be noted that the apertures **36** going through the front face **34** of the housing **12** are positioned such that they are intersecting with the curved inner wall **16** of the housing **12**. A removal key **70** is also shown in the figures. The removal key **70** includes key arm **72** and an inlet **76** connected to a base **74**. The arm **72** and inlet **76** are configured to match or at least be used with the aperture **36** and wall **37** of the housing **12**. As noted, the glue board can include an adhesive on both sides thereof. When such a configuration is used, there may be increased difficulty to remove the glue board from the inner wall **16** of the housing **12**. In such a situation, the removal key **70** could be utilized with the housing **12** to aid and urge the removal thereof. For example, the key arm **72** are lined with the inlet **76** in the wall **37** as well as with the aperture **76** of the housing **12**. Insertion of the arm **72** into the aperture **36** will contact the arms with the interior portion of the glue board adjacent the wall **16** thereof. This will urge disconnection of the glue board from the wall **16**, which will aid in the removal of the glue board by creating an initial separation thereof, at which time a user can continue the separation thereof, such as by continuing the peeling of the glue board from the inner wall **16** of the housing **12**. In addition, even when the glue board does not have the adhesive on the wall side thereof, the key **70** can still be used to aid in removing the glue board **50** from the housing **12**, such as by releasing or separating the glue board from the ledge or notch **30**, which will make it easier to remove the glue board from the interior side of the housing **12**.

(46) As noted herein, a glue board **50** has been described as being used with the housing **12**. The glue board **50** is utilized to provide an adhesive portion to capture or catch insects. The glue board and/or the device **10** can also include a pesticide or other killing agent to both capture and kill insects passing therethrough. While it is to be appreciated that generally any type of glue board capable of capturing and/or killing insects can be used, FIGS. **13-15** and **17** provide an exemplary embodiment of a glue board for use with the housing and according to aspects and/or embodiments of the present invention. As best shown in FIG. **17**, the glue board **50** is provided in schematic form. It should be noted that FIG. **17** actually shows two glue boards, which will be designated by A and B identifiers in addition to the numerals thereof. For example, as shown in FIG. **17**, a perforated line **66** is provided which allows for two boards, **50**A and **50**B, to be made at the same time. While not needed in all instances, it can aid in the manufacture thereof. However, as will be understood, a single board can be made and utilized as well. The exemplary glue board **50** is used with the housing **12** will be shown with respect to one of the sides as shown in FIG. **17**. For example, it should be appreciated that the left side of FIG. **17** is notified as glue board **50**A. It should be appreciated that **50**B will be generally the same and opposite in a mirrored fashion thereof. The glue board **50**A includes a base portion **52**A. The base portion can be cardboard, paper, polymer, wood, steel, or generally any other material for holding the components of the board **50**. The base portion **52**A includes a first section **54**A, which includes a glue or other adhesive positioned and on the base **52**A. Adjacent the first section **54**A is a second section **56**A. The second section **56**A is a no glue area, at least on one side thereon. It should be appreciated that the second section **56**A having no glue or adhesive is generally at or near a periphery **58**A thereof and is also sized smaller than the first section **54**A including the glue or adhesive. As will be understood with respect to the installation and servicing of the device **10**, having the first section **54**A with the adhesive or glue, and an adjacent section **56**A at a periphery **58**A and having generally no glue or adhesive will allow for the easy installation and removal of the glue board **50**A with respect to the housing **12**. Likewise, the same features are found in glue board **50**B, including the base **52**B, the first section **54**B with the adhesive, and the second section **56**B being substantially free of adhesive at and near the periphery **58**B thereof.

(47) In addition to the glue board itself, a protective cover **60**, which is shown as **60**A and **60**B in FIG. **17**, can be provided. The protective cover comprises paper, such as wax paper or other material that can be temporarily adhered and removed without substantial ripping or tearing, which may also be referred to as a pull tab, can cover the adhesive first section **54**A until such time that the glue board **50**A has been installed with respect to a housing **12**, to protect the adhesive portion until such time it will be positioned and utilized in the field, such as when it is positioned at a surface. However, it should also be appreciated that the pull tab or cover **60**A may not be required in all embodiments or aspects thereof.
(48) The no glue second section **56**A acts as the glue keep out which may be required or otherwise useful for production rollers to pull the glue board down a production line. This also functions as a mounting feature, as will be understood, useful to install the glue board **50** quickly and effortlessly with the pull tab or protective cover **60**A remaining free of the plastic undercut.
(49) Such as glue board is shown in FIGS. **13** and **14** after production and after being separated via the perforation line **66**. Therefore, as shown in FIGS. **13**, **14** only one portion of the dual made glue board is included. As noted, the glue board **50** includes the base **52**. Furthermore, the glue board will include a first section **54** having the glue or other adhesive and a second section **56**, which is substantially free from glue or adhesive. This is further illustrated by the protective cover **60** being adhered to the first section **54**, while being generally separated from the second section **56**. This may be due in part to the second section **56** not having a glue or other adhesive thereat, which will prevent or preclude the cover **60** attaching thereto. Still further, it should be appreciated that the second section **56** is generally at or near perforated **58** of the base **52**. As will be understood, this will aid in the installation and servicing of the glue board for use with the housing **12** of the insect capture device **10**.
(50) As best shown in FIGS. **16**A-**16**D, the installation procedure of the glue board **50** will be described. While the installation shown in the figures is provided, it should be appreciated that it is not limiting, and other manners of installation are to be envisioned and included. However, the figures and description are included for purposes of disclosure and to show the advantages of the glue board having a section of glue adjacent a section of non-glue for purposes of use with the housing **12** of the insect device **10**.
(51) Therefore, as shown in FIGS. **16**A, the glue board **50** having the protective cover **60** is inserted in the second notch or ledge **30** of the housing **12** at the inner side **16** thereof. This is shown generally by the arrow **61**. It will be noted that the portion inserted into the ledge **30** generally includes the section **54** of the glue board **50** having the adhesive and being covered at least in part by the protective cover **60**. Next, the opposite edge **58** of the glue board **50** which is the second section **56** having substantially no glue or adhesive thereat is positioned into and in connection with the first ledge or notch **28** at the inner side **16** of the housing **12**. See, e.g., FIG. **16**B, with the arrow **63** showing the positioning of the board into the notch. As shown in FIG. **16**B, both sides of the glue board are positioned generally in the notches **28**, **30**. However, it is also noted in FIG. **16**B that the glue board is not necessarily conforming to the shape of the raised portion **22** of the housing **12**.
(52) This can be easily rectified such as shown in FIG. **16**C, where the glue board is manipulated to conform generally to the shape of the raised position **22** of the housing at the inner portion **16** thereof. As shown, the glue board **50** is initially shaped in a convex manner, such as is shown by the dash line **50**′ in FIG. **16**C. Pressure can be applied, such as in the direction of the arrow **65**, which moves the glue board **50** into generally conformity with the shape of the housing **12**, which is shown by the solid line **50** in FIG. **16**C. In addition, as has been disclosed, the glue board can include an adhesive in full or part on the housing side to aid in conforming of the shape and maintaining the position of the glue board relative to the housing, but it is not required or included in all embodiments or aspects disclosed.
(53) Next, as shown best in FIG. **16**D, once the glue board **50** has been positioned between the notches **28**, **30** and has been manipulated to generally conform to the shape of the raised portion of

the housing, before the device can be used, the adhesive of the first section **54** of the glue board **50** needs to be exposed. Therefore, the protective cover can be peeled from the adhesive first section **54** of the glue board **50**. This is shown generally by the direction of the arrow **64** in FIG. **16**D. the paper will generally pull off of the glue board simply and pulling in a single motion due to the composition of the protective cover and the glue board. This will allow for a full exposure of the adhesive first section **54** positioned away from the housing **12** and towards the surface **62** such that the full adhesive **54** of the glue board **50** can be used to attract and capture any insect passing therethrough or thereunder. The completed installation is shown in FIG. **15**, where the glue board **50** has been positioned between the notches **28**, **30**. It should be noted that FIG. **15** does include the protective cover or paper **60** in conjunction and connected to the glue board **50**. However, once the housing **12** is placed for final position, the cover **60** can simply be peeled from the glue board **50** such as in the arrow **64** shown as FIG. **16**D to remove and expose the adhesive away from the housing for use in catching and capturing an insect passing thereby.
(54) Therefore, a combination of an insect capture device including a housing and a glue board has been shown and described. It is noted that combination can include numerous variations and need not include every single aspect and/or feature as disclosed herein. Still further, it should be appreciated that while some variations have been shown and described, any of the variations of any of the embodiments can be combined to create new embodiments or aspects, including those not explicitly shown or described herein. Still further, it should be noted that the combination of the glue board with the sectioned adhesive and non-adhesive being adjacent to one another for use with a housing having a raised portion to allow for insects to pass thereto and to be exposed to the adhesive section will provide an improved insect device for monitoring, capturing, trapping and/or killing insects passing thereto.
(55) From the foregoing, it can be seen that the invention accomplishes at least all of the stated objectives.

Claims

1. In combination, an insect capture device and an adhesive member, comprising: the insect capture device comprising a housing having a raised portion to create a gap between the raised portion and a surface; and the adhesive member comprising a first section comprising an adhesive and a second section adjacent the first section and being substantially free from any adhesive; wherein the adhesive member is positioned at an underside of the raised portion of the housing of the insect capture device with the first section of the adhesive member facing the gap; wherein the housing comprises first and second support members on opposed ends of the raised portion; wherein the first and second support members are configured to touch the surface when the combination is in use; and wherein when the first and second support members touch the surface, the first and second support members form a partial enclosure between the raised portion and the surface such that the first and second support members do not create a sealed enclosure entirely surrounding the gap; wherein the housing of the insect capture device further comprises at least one notch on either of the first and second support members, and wherein the adhesive member is positioned at least partially in the at least one notch.
2. The combination of claim 1, wherein the raised portion of the housing comprises a curved shape.
3. The combination of claim 1, wherein the at least one notch comprises first and second notches opposite one another on the first and second support members to receive opposing edges of the adhesive member.
4. The combination of claim 3, wherein the second section of the adhesive member is configured to be positioned in one of the first or second notches.
5. The combination of claim 1, wherein the adhesive member is a glue board, and the adhesive is a glue.

6. The combination of claim 1, wherein the adhesive member further comprises a protective cover covering substantially both the first and second sections of the adhesive member.
7. The combination of claim 1, wherein the insect capture device further comprises a base connected to the housing.
8. The combination of claim 7, wherein the base comprises at least one aperture and the housing comprises at least one protruding member, and the at least one protruding member temporarily secured to the at least one aperture to connect the base to the housing.
9. An insect station, comprising: a housing comprising an inner portion and an outer portion, the inner portion including a support portion and a raised portion, wherein the support portion comprises first and second support members on opposed ends of the raised portion and said raised portion creating a gap between the first and second support members; a glue board positioned at the inner portion of the housing, said glue board comprising a first section comprising an adhesive and a second section adjacent the first section and being substantially free from any adhesive; and first and second notches on the first and second support members respectively, and wherein the glue board positioned at least partially in the first and second notches; wherein the glue board is positioned to expose the adhesive of the first section to face the gap; wherein the first and second support members are configured to touch a surface when the insect station is in use; and wherein when the first and second support members touch the surface, the first and second support members form a partial enclosure between the raised portion and the surface such that the first and second support members do not create a sealed enclosure entirely surrounding the gap.
10. The insect station of claim 9, wherein the raised portion comprises a curved shape.
11. The insect station of claim 9, wherein the second section of the glue board being substantially free from adhesive is positioned in the first or second notch.
12. The insect station of claim 9, wherein the glue board further comprises a protective cover releasably attached to the adhesive of the first section and covering both the first and second sections of the glue board.
13. The insect station of claim 9, further comprising a base operatively attached to the housing, the base configured to be affixed to the surface.
14. A system for capturing insects, comprising: a housing with a gap between the housing and a mounting surface; and a glue board releasably attached to the housing, the glue board comprising a first section including an adhesive for capturing insects and a second section along at least a periphery of the glue board to interact with the housing to hold the glue board in place with the adhesive of the first section positioned towards the gap; wherein the housing comprises first and second support members on opposed ends of the gap; and wherein the first and second support members are configured to touch the mounting surface when the system is in use; and wherein when the first and second support members touch the mounting surface, the first and second support members form a partial enclosure between the housing and the mounting surface such that the first and second support members do not create a sealed enclosure entirely surrounding the gap; wherein the housing of the system further comprises at least one notch on either of the first and second support members, and wherein the glue board is positioned at least partially in the at least one notch.
15. The system of claim 14, wherein the second section of the glue board along one or more edges of the glue board, and the glue board further comprising a protective cover releasably attached to the first section and substantially covering both the first and second sections thereof.